Optimum and Adaptive Signal Processing Systems



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Content

- Random Variables and Signals
  - Random Variables
  - Random Vectors, Multivariate Gaussian Distribution
  - Random Signals (Stochastic Processes)
- Basics of Estimation Theory with Applications in Signal Processing
  - Classical Estimation: CRLB, MVU, BLUE, ML, LS, Estimation of ACS and PSD of WSS processes
  - Bayesian Estimation: MMSE, MAP, LMMSE
- Optimum Filters: Wiener and Least Squares Filters
  - Wiener Filters
  - Least Squares Filters
  - Applications: (Inverse) System Identification, Linear Prediction (e.g. for speech), Noise Cancellation
- Adaptive Filters (and Applications)
  - LMS
  - RLS
- Kalman Filter and Extended Kalman Filter (with Applications)

Content
Random Variables and Signals

- Random Variables
  - Definition of random variables
  - Distribution functions (CDF, PDF, PMF)
  - Expectation and moments
  - Uniform distribution and Gaussian distribution
- Random Vectors, Multivariate Gaussian Distribution
- Random Signals (Stochastic Processes)

References

[1] S. Kay, *Intuitive Probability and Random Processes using MATLAB*, Springer, 2005.

[2] H. Hsu, *Schaum's outlines: Probability, Random Variables, and Random Processes*, Second Edition, Schaum's Outline Series, Mc Graw Hill, 2011.

[3] G. Manolakis, V.K. Ingle, S.M. Kogon, *Statistical and Adaptive Signal Processing*, Artech House, 2005.

[4] D. Ch. von Grünigen, *Digitale Signalverarbeitung*, Hanser, München, 2008.

Random Variables

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Random experiments, probability:

a) *The set of all possible outcomes of a random experiment is called the sample space, and it is denoted by S. The elements $s \in S$ are the possible outcomes of the random experiment which are also called sample points.*

b) *Any subset of the sample space S is called an event.*

c) *Events that contain only a single outcome such as $\{s\}$ are called simple events or elementary events.*

d) *We say that an event A occurs if the outcome s is an element of the defining set of that event A, i.e. if $s \in A$.*

e) *Let A be a particular event. The probability of the event A is denoted by $P(A)$.*

Random Variables

Consider a random experiment with sample space S.

a) *A random variable $X(s)$ is a single-valued real function*

$$X : S \rightarrow \mathbb{R}$$

that assigns a real number, called the value of $X(s)$, to each sample point $s \in S$. Frequently, we use a single letter X for this function in place of $X(s)$, and use RV to denote the random variable.

b) *The collection of all numbers*

$$S_X := \{X(s) | s \in S\}$$

which is a subset of $\mathbb{R}$ is called the range of the RV.

Random Variables

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Example: Assigning 0 and 1 to head and tail of a coin

→ more compact description via RV

Comments

- Note that the terminology used here is traditional. Clearly a random variable is not a variable at all in the usual sense, but it is a function.
- For numerically valued random experiments in which $s \in \mathbb{R}$, we can still use the random variable approach if we e.g. define $X(s) = s$ for all s (e.g. dice).

Notation

- X…random variable,
- x…realization of the RV or value that was actually observed after the random experiment

Random Variables
CDF, PDF, PMF

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Cumulative distribution function (CDF):

- *We define the event $(X \leq x)$ as the set of all sample points (possible outcomes) for which $X(s) \leq x$:*

$$(X \leq x) := \{s \in S | \, X(s) \leq x\}$$

- *The function*

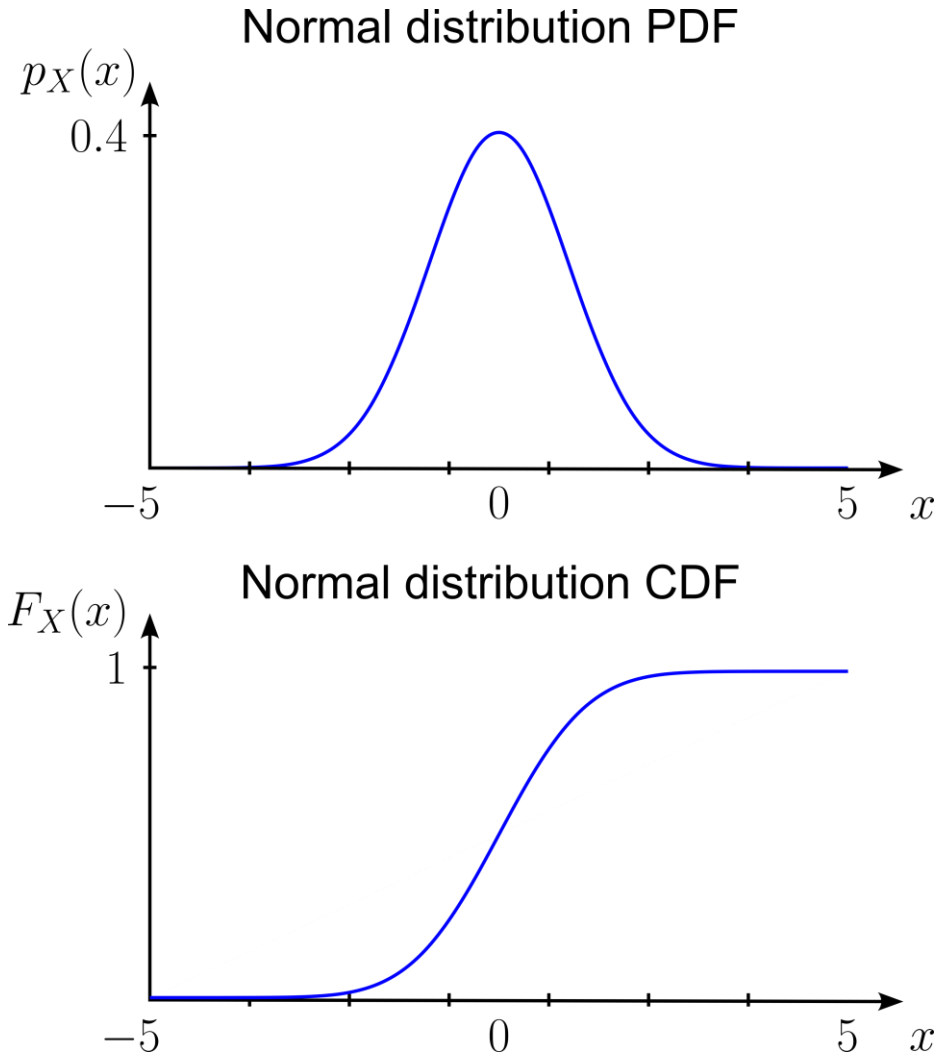
$$F_X(x) := P(X \leq x) \qquad -\infty < x < \infty$$

is called the cumulative distribution function (CDF) of the random variable X.

Most of the information about a random experiment described by the RV X is determined by the behavior of the CDF.

Random Variables
CDF, PDF, PMF

CDF example for normal distribution:



Random Variables
CDF, PDF, PMF

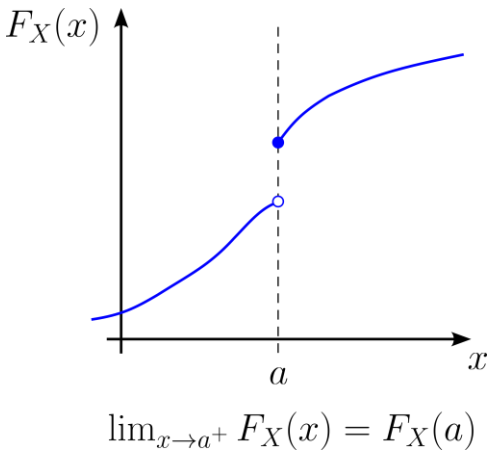
- Several properties of the CDF follow directly from its definition
  - $0 \leq F_X(x) \leq 1$
  - $F_X(x_1) \leq F_X(x_2)$ for $x_1 < x_2$ ($F_X(x)$ is monotonically increasing)
  - $\lim_{x\to\infty} F_X(x) = 1, \lim_{x\to-\infty} F_X(x) = 0$
  - $\lim_{x\to a^+} F_X(x) = F_X(a)$ ($F_X(x)$ is continuous on the right)

- Determination of probabilities from the CDF:

$$P(X > a) = 1 - F_X(a)$$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

$$P(X < b) = F_X(b^-)$$



Discrete Random Variables and Probability Mass Function (PMF)

- *A random variable X is called a discrete RV if its range S_X contains a finite or countably infinite number of points. For a discrete RV the CDF $F_X(x)$ is a staircase function with at most a countable number of jumps.*

- *Suppose that the jumps in $F_X(x)$ of a discrete RV X occur at the points $x_1, x_2, \ldots$ where the sequence may be either finite or countably infinite, and we assume $x_i < x_j$ if $i < j$. Then*

$$F_X(x_i) - F_X(x_{i-1}) = P(X \leq x_i) - P(X \leq x_{i-1}) = P(X = x_i).$$

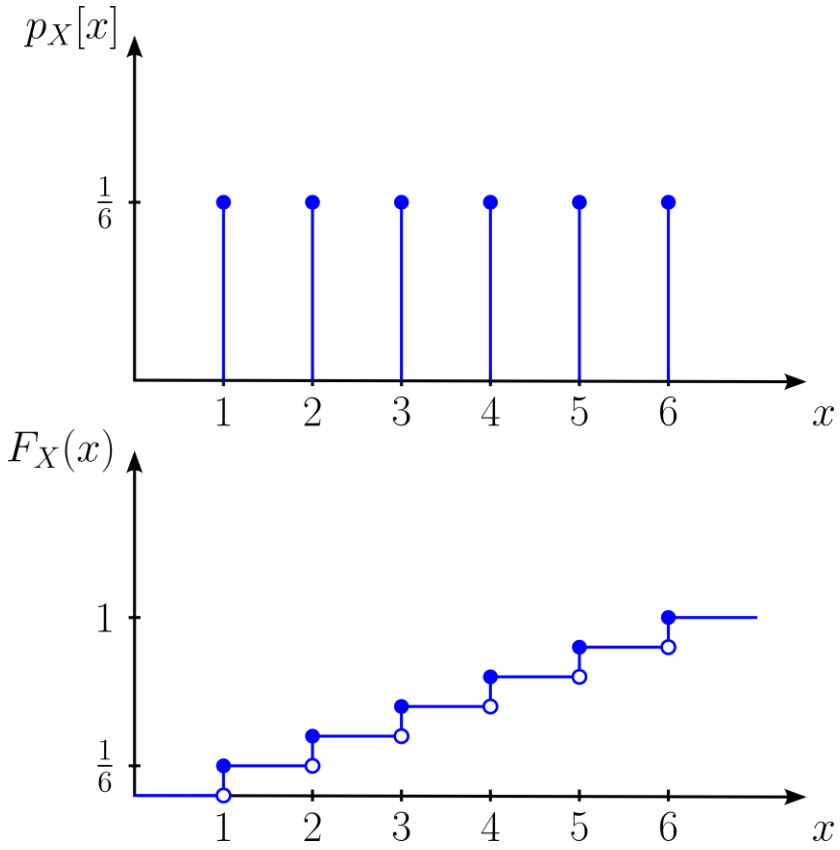
The function

$$p_X[x] = P(X = x)$$

is called the probability mass function (PMF) of the discrete RV X.

Random Variables
CDF, PDF, PMF

Example for PMF and corresponding CDF of throwing a dice:



This distribution is called the *discrete uniform distribution*.

Random Variables
CDF, PDF, PMF

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Discrete Random Variables and Probability Mass Function (PMF)

- Properties of PMFs:
  - $0 \leq p_X[x_i] \leq 1 \qquad i = 1,2,\ldots$
  - $p_X[x] = 0 \qquad$ for $x \neq x_i \;\; (i = 1, 2, \ldots)$
  - $\sum_i p_X[x_i] = 1$

- Relationships between CDF and PMF:

$$F_X(x) = P(X \leq x) = \sum_{i:x_i \leq x} p_X[x_i]$$

$$p_X[x_i] = F_X(x_i) - F_X(x_{i-1})$$

Continuous Random Variables and Probability Density Function

- *A random variable X is called a continuous RV if its range S_X consists of an infinite and uncountable number of outcomes (containing at least one interval – either finite or infinite - of real numbers).*

- For a continuous RV with CDF $F_X(x)$ the derivative $\mathrm{d}F_X(x)/\mathrm{d}x$ exists everywhere except at possibly a finite number of points.

- *For continuous RVs the probability density function (PDF) $p_X(x)$ is defined as a formal derivative of the CDF*

$$p_X(x) := \frac{\mathrm{d}F_X(x)}{\mathrm{d}x}$$ [1)]

1) *at the points where the derivative exists.*

Probability Density Function (PDF)

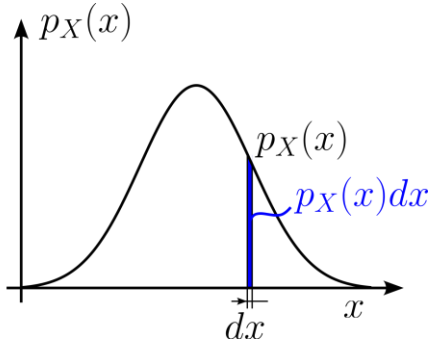
- Note that $p_X(x)$ is not the probability, but the probability density:

$$p_X(x) = \frac{\mathrm{d}F_X(x)}{\mathrm{d}x} = \lim_{\varepsilon \to 0} \frac{F_X(x+\varepsilon/2) - F_X(x-\varepsilon/2)}{\varepsilon}$$

$$\cong \frac{F_X(x+\mathrm{d}x/2) - F_X(x-\mathrm{d}x/2)}{\mathrm{d}x} = \frac{P(x-\mathrm{d}x/2 < X \leq x+\mathrm{d}x/2)}{\mathrm{d}x}$$

or

$$\boxed{p_X(x)\mathrm{d}x = P(x-\mathrm{d}x/2 < X \leq x+\mathrm{d}x/2)}$$



- If X is a continuous RV, then $P(X = x) = 0$. Note that this is an example of an event with probability 0 that is not necessarily the impossible event.

Random Variables
CDF, PDF, PMF

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Probability Density Function (PDF)

- Integrating both sides of $p_X(x) = \mathrm{d}F_X(x)/\mathrm{d}x$ we obtain

$$F_X(x) = \int_{-\infty}^{x} p_X(u)\mathrm{d}u$$

- Properties of PDFs:
  - $p_X(x) \geq 0$
  - $\int_{-\infty}^{\infty} p_X(x)dx = 1$
  - $$P(a < X \leq b) = F_X(b) - F_X(a) = \int_{a}^{b} p_X(x)\mathrm{d}x$$
  - $p_X(x)$ is piecewise continuous

Random Variables
CDF, PDF, PMF

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Probability Density Function (PDF)

- For continuous RVs we have

$$
\begin{aligned}
P(a < X \leq b) &= P(a \leq X < b) \\
&= P(a < X < b) \\
&= P(a \leq X \leq b) \\
&= F_X(b) - F_X(a) = \int_a^b p_X(x)\mathrm{d}x\,.
\end{aligned}
$$

Random Variables
Statistical Averages

- To completely characterize a random variable, we have to know its probability density function.
- In practice, it is desirable to summarize some of the key aspects of a density function by using a few numbers rather than to specify the entire density function.
- These numbers, which are called *statistical averages* or *moments*, are evaluated by using the mathematical expectation operation.
- Although density functions are needed to theoretically compute moments, in practice, moments are typically easily estimated without the explicit knowledge of density functions.

Random Variables
Statistical Averages

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Mathematical expectation

- *The expected or mean value of a discrete RV X is defined by*

$$E[X] = \mu_X := \sum_{i=-\infty}^{\infty} x_i p_X[x_i] .$$

- *For continuous RVs the mean is defined by*

$$E[X] = \mu_X := \int_{-\infty}^{\infty} x p_X(x) \mathrm{d}x .$$

- Simple rule:

$$E[X] = \sum_i \begin{pmatrix} (\text{Possible value of the random variable}) \times \\ (\text{Probability that the random variable takes on this value}) \end{pmatrix}$$

Mathematical expectation

- The expectation operation computes a statistical average by using the density $p_X(x)$ as a weighting function.
- Hence, the mean μ_X can be regarded as the "center of gravity" of the density $p_X(x)$.
- Important properties of the expectation operator:
  - ☐ Linearity: $E[\alpha X] = \alpha E[X]$ — α … constant
  - ☐ Composition: $E[g(X)] = \int_{-\infty}^{\infty} g(x) p_X(x) \mathrm{d}x$ — $g(.)$ … an arbitrary function

Mathematical expectation

- The latter result is quite important. Frequently we work with a function $Y = g(X)$ of a random variable X. In principle we could calculate the PDF (or PMF) of Y and derive the expectation of Y per definition

$$E[Y] = \int_{-\infty}^{\infty} y p_Y(y) \mathrm{d}y \,.$$

- The above composition property tells us that the knowledge of the PDF of Y is not required to calculate its mean, we can also average over $g(X)$ with respect to the PDF of X:

$$E[Y] = E[g(X)] = \int_{-\infty}^{\infty} g(x) p_X(x) \mathrm{d}x$$

Moments

- Using the expectation operation and the rule of composition we can define various moments of the random variable X that describe certain useful aspects of the density function.

- *Let* $g(X) = X^m$. *Then*

$$r_X^{(m)} = \int_{-\infty}^{\infty} x^m p_X(x)\mathrm{d}x$$

is called the mth*-order moment of* X. *In particular,* $r_X^{(0)} = 1$, *and the first-order moment* $r_X^{(1)} = \mu_X$.

- The second-order moment $r_X^{(2)} = E[X^2]$ is called the *mean-squared value*, and it plays an important role in estimation theory.

Random Variables
Statistical Averages

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Moments

- *Let* $g(X) = (X - \mu_X)^m$, *then*

$$\gamma_X^{(m)} = \int_{-\infty}^{\infty} (x - \mu_X)^m p_X(x) \mathrm{d}x$$

is called the mth*-order central moment of* X.

- Clearly, a random variable's moments and central moments are identical if its mean is zero.

- *The second central moment is of considerable importance and is called the variance of* X*, denoted by* σ_X^2*. Thus*

$$\mathrm{var}[X] = \sigma_X^2 = E[(X - E[X])^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 p_X(x) \mathrm{d}x \,.$$

Moments

- The quantity $\sigma_X = \sqrt{\mathrm{var}(X)}$ is called the *standard deviation* of X and is a measure of the spread (or dispersion) of the observed values of X around its mean μ_X.

- Properties of the variance:

$$\mathrm{var}[X] = E[X^2] - (E[X])^2 \quad \text{or} \quad \sigma_X^2 = r_X^{(2)} - \mu_X^2$$

$$\mathrm{var}[\alpha X + \beta] = \alpha^2 \mathrm{var}[X]$$

α, β … constants

- ***Chebyshev's inequality:*** $P(|X - \mu_X| \geq k \cdot \sigma_X) \leq \frac{1}{k^2}$

This means, that regardless of the shape of $p_X(x)$, the probability that the random variable X deviates from its mean by more than k times its standard deviation is less than or equal $1/k^2$.

Random Variables
Some Useful Random Variables

- Random variable *models* are needed to describe or approximate complex physical phenomena using simple parameters.

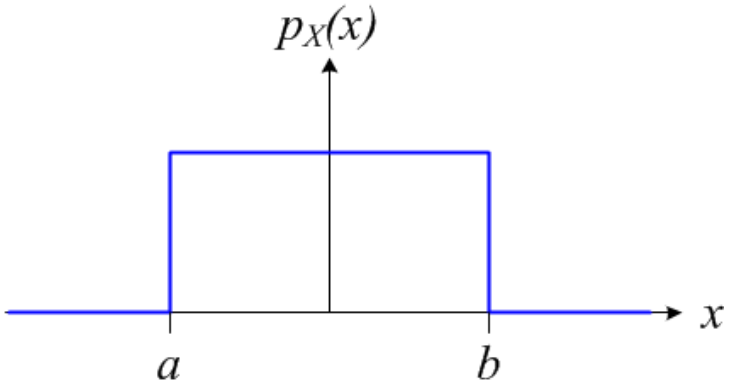
- For example, the random phase of a sinusoidal carrier is frequently modeled by a uniformly distributed random variable.

Uniformly distributed random variable

- This is an appropriate model in situations in which random outcomes are „equally likely.“

- The PDF of a uniformly distributed continuous RV X is defined by

$$p_X(x) = \begin{cases} \dfrac{1}{b-a} & a \le x \le b \\ 0 & \text{elsewhere} \end{cases}$$



where $a < b$ are specified parameters.

Uniformly distributed random variable

- The corresponding CDF is given by

$$F_X(x) = \int_{-\infty}^{x} p_X(u)\mathrm{d}u = \begin{cases} 0 & x < a \\ \dfrac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

- The mean and the variance of this random variable are given by

$$\mu_X = \frac{a+b}{2}$$
$$\sigma_X^2 = \frac{(b-a)^2}{12}$$

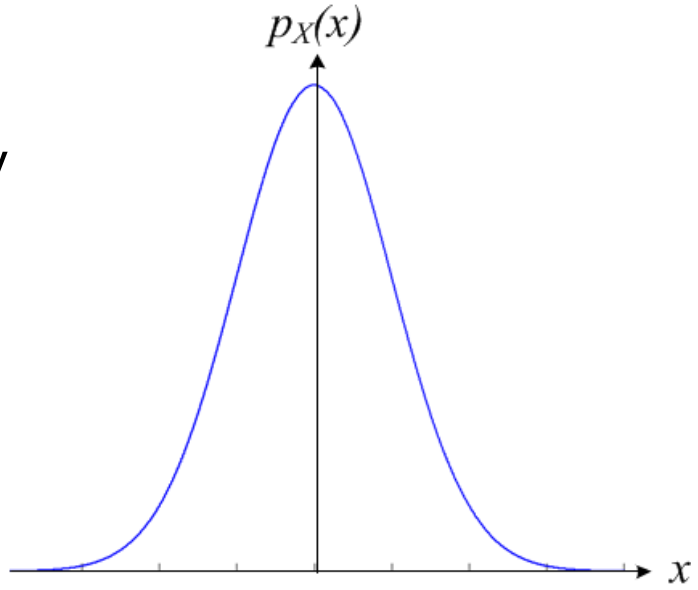
Random Variables
Some Useful Random Variables

Normal distributed random variable

- In many applications this is the most useful and convenient model, as we shall see later.
- It is also known as a Gaussian random variable, and we will use both terms interchangeably.
- The PDF of a normally distributed RV X with mean μ_X and variance σ_X^2 is given by

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2\right]$$

where $-\infty < \mu_X < \infty$ and $\sigma_X \geq 0$.



Random Variables
Some Useful Random Variables

Normal random variable

- The PDF of a Gaussian random variable is completely described by its mean μ_X and variance σ_X^2 and is denoted by $\mathcal{N}(\mu_X, \sigma_X^2)$.
- We note that all higher-order moments of a normal random variable can be determined in terms of the first two moments, that is,

$$\gamma_X^{(m)} = E\{[X - \mu_X]^m\} = \begin{cases} 1 \cdot 3 \cdot 5 \cdots (m-1)\sigma_X^m & , \quad \text{if } m \text{ even} \\ 0 & , \quad \text{if } m \text{ odd} \end{cases}$$

- The higher moments do not provide any additional information about the Gaussian density function.

Content
Random Variables and Signals

- Random Variables
- Random Vectors, Multivariate Gaussian Distribution
  - Definition
  - Joint CDF, joint PMF, conditional PMF
  - Joint PDF, conditional PDF
  - Covariance and correlation coefficient
  - Mean vector and covariance matrix
  - The multivariate Gaussian distribution
- Random Signals (Stochastic Processes)

Random Vectors:

- In many applications, a group of signal observations can be modeled as a collection of random variables that can be grouped to form a random vector.
- Random signals of finite duration can e.g. be modeled by random vectors.
- This is an extension of the concept of a random variable and generalizes many scalar quantities to vectors and matrices.
- A real-valued vector containing n random variables

$$\mathbf{X}(s) = [X_1(s), X_2(s), \ldots, X_n(s)]^T$$

is called a random n-vector or a random vector when the dimensionality is clear from the context.

Random Vectors
CDF, PMF, PDF

JKU

- For clarity reasons we define the CDFs, PMFs and PDFs for 2-dimensional random vectors $\mathbf{X} = [X, Y]^T$. The extension to n-dimensional vectors is straight forward, we refer to the literature for details.

Distributions and density functions

- *A 2-dimensional random vector is completely characterized by its joint cumulative distribution function, which is defined by*

$$F_{X,Y}(x, y) := P(X \leq x, Y \leq y)$$

and is often written as

$$F_{\mathbf{X}}(\mathbf{x}) = P(\mathbf{X} \leq \mathbf{x})$$

for convenience.

Random Vectors
CDF, PMF, PDF

JKU

Joint Probability Mass Function

- *Let* $\mathbf{X} = [X, Y]^T$ *be a 2-dimensional random vector, and let* $[X, Y]^T$ *take on the values* $[x_i, y_j]^T$ *for a certain allowable set of integers* i *and* j**.** *The function* $p_{\mathbf{X}}[\boldsymbol{x}]$

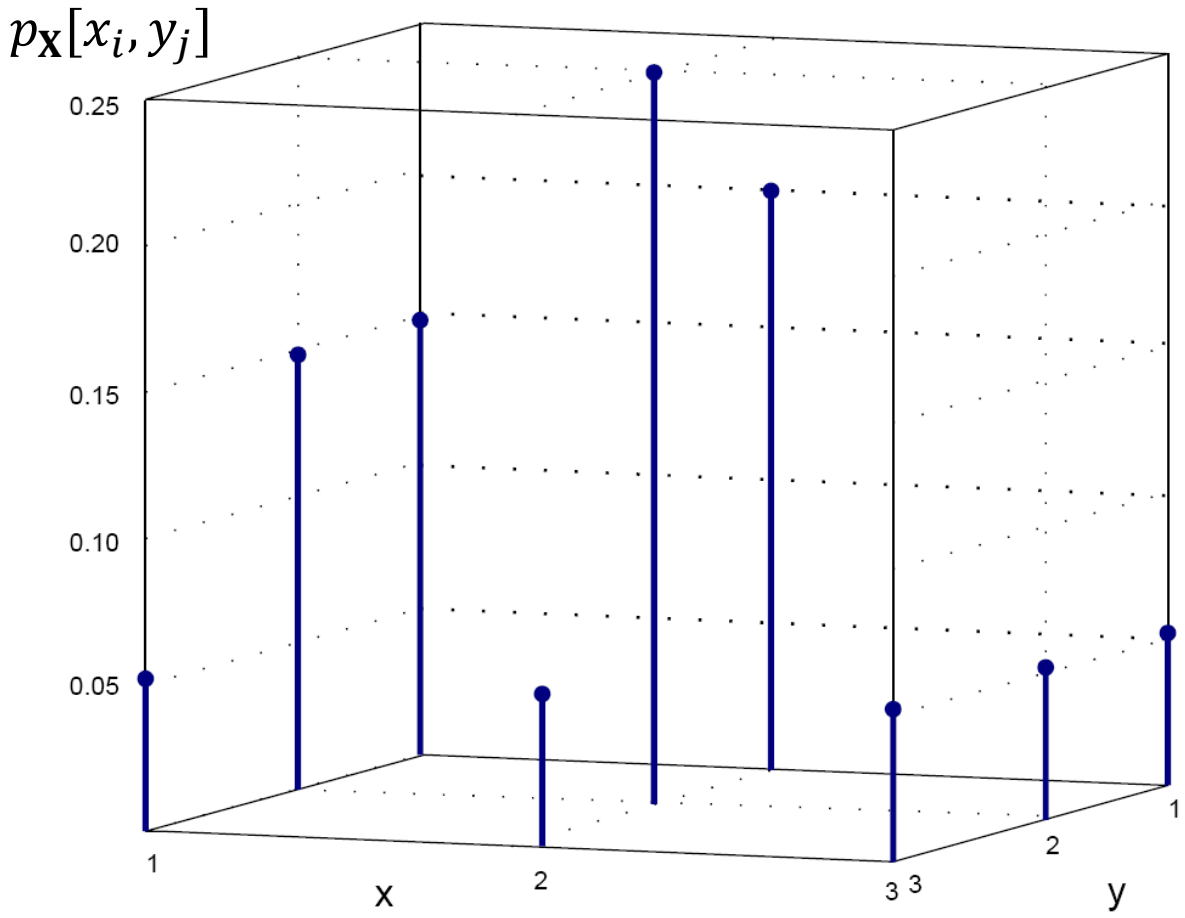
$$p_{\mathbf{X}}[\mathbf{x}] = p_{X,Y}[x, y] = P(X = x, Y = y)$$

is called the joint probability mass function (joint PMF) of $\mathbf{X}$**.**

- Properties:
  - $0 \leq p_{\mathbf{X}}[x_i, y_j] \leq 1$
  - $\sum_i \sum_j p_{\mathbf{X}}[x_i, y_j] = 1$

Random Vectors
CDF, PMF, PDF

- Example of a joint PMF:



Joint Probability Mass Function

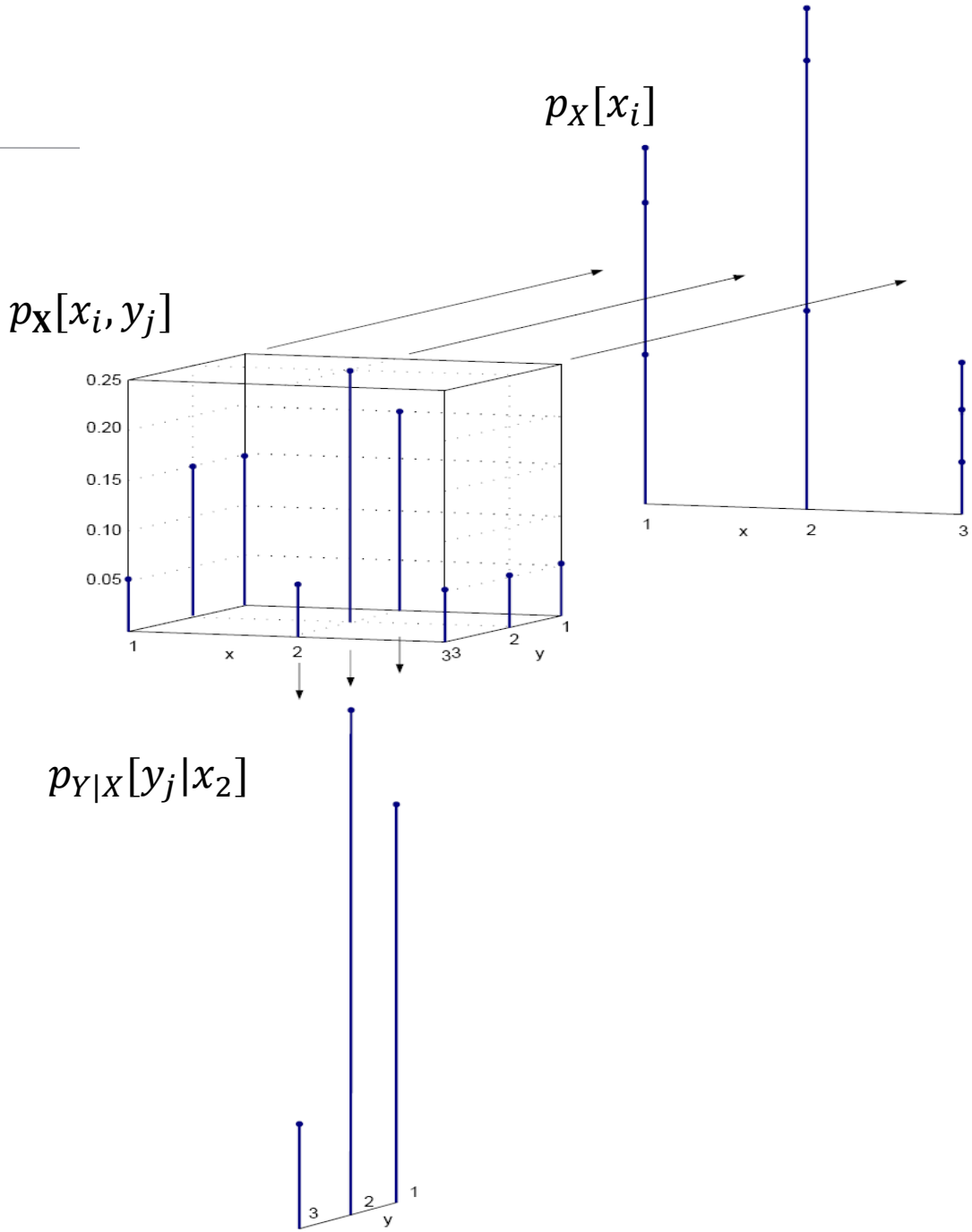
- Properties:
  - The probability of an event A is simply given by summing $p_{\mathbf{X}}[x_i, y_j]$ over all points (x_i, y_j) in the range space corresponding to the event A.
  - Relationship between joint CDF and joint PMF:

$$F_{\mathbf{X}}(x, y) = \sum_{i:x_i \le x} \sum_{j:y_j \le y} p_{\mathbf{X}}[x_i, y_j]$$

  - Marginal PMFs:

$$p_X[x_i] = \sum_j p_{\mathbf{X}}[x_i, y_j]$$

$$p_Y[y_j] = \sum_i p_{\mathbf{X}}[x_i, y_j]$$


$p_X[x_i]$
$p_{\mathbf{X}}[x_i, y_j]$
0.25
0.20
0.15
0.10
0.05
1
x
2
3
3
2
y
1
1
x
2
3
$p_{Y|X}[y_j|x_2]$
3
2
y
1

Joint Probability Mass Function

- *Two discrete random variables are defined to be statistically independent, if*

$$p_{\mathbf{X}}[x, y] = p_X[x] p_Y[y].$$

- This is equivalent to

$$F_{\mathbf{X}}(x, y) = F_X(x) F_Y(y).$$

Conditional Probability Mass Function

- The *conditional* PMF of Y, given that $X = x_i$, is defined as

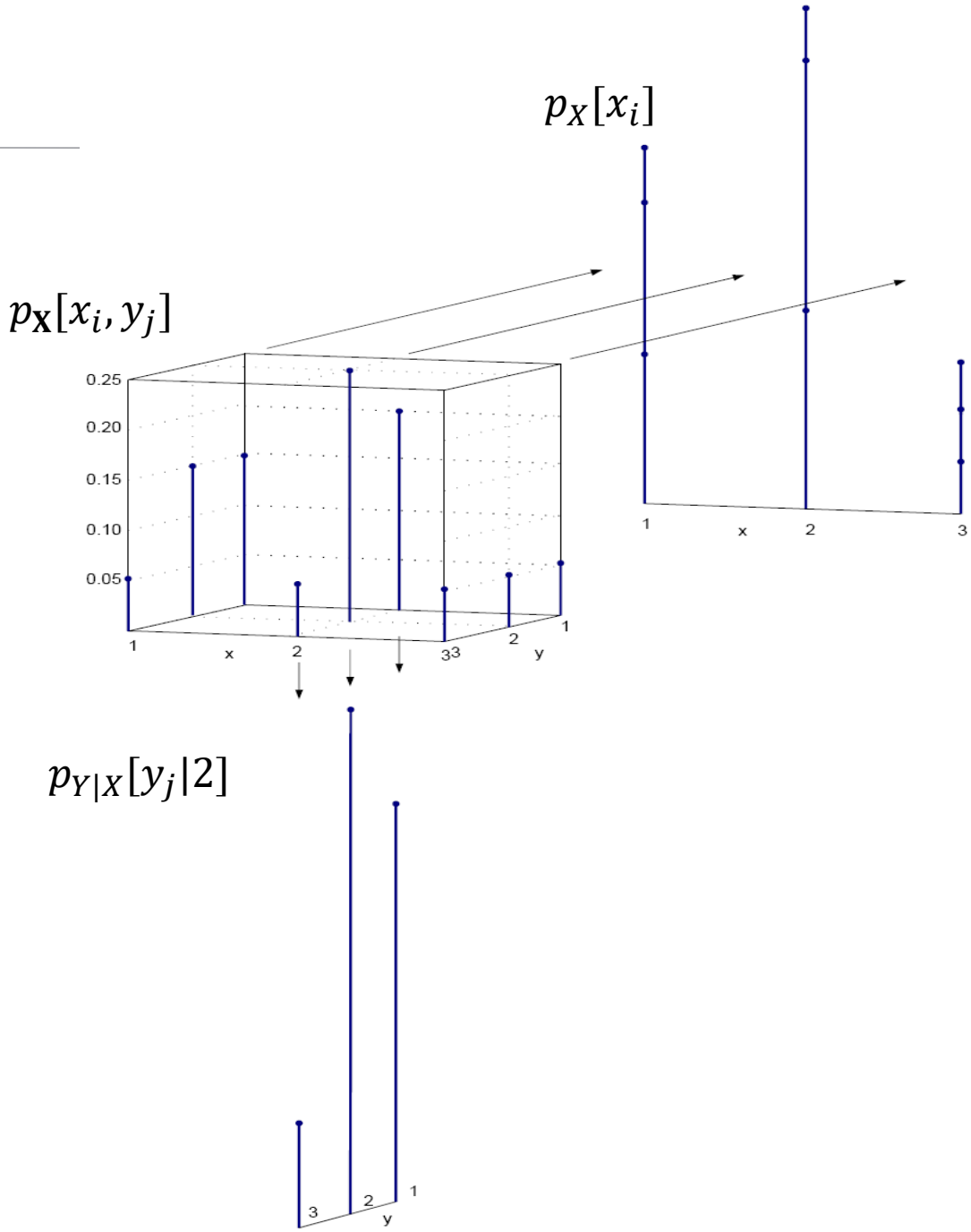
$$p_{Y|X}[y_j|x_i] := P(Y = y_j|X = x_i)$$

- and is given by

$$p_{Y|X}[y_j|x_i] = \frac{p_{\mathbf{X}}[x_i, y_j]}{p_X[x_i]} \qquad p_X[x_i] > 0$$

- The corresponding definition also holds for $p_{X|Y}[x_i|y_j]$.

- Notice that if X and Y are independent, then $p_{Y|X}[y_j|x_i] = p_Y[y_j]$.


$p_X[x_i]$
$p_{\mathbf{X}}[x_i, y_j]$
0.25
0.20
0.15
0.10
0.05
1
x
2
3
3
2
y
1
1
x
2
3
$p_{Y|X}[y_j|2]$
3
2
y
1

Conditional Mean and Conditional Variance

- Conditional expectations are defined in the same way as for marginal PMFs.
- The *conditional mean* of Y, given that $X = x_i$, is defined as

$$\mu_{Y|X} = E_{Y|X}[Y|x_i] = \sum_j y_j p_{Y|X}[y_j|x_i]$$

- Note that the conditional mean is a function of x_i.
- The *conditional variance* is defined as

$$\begin{aligned}\mathrm{var}[Y|x_i] &= E_{Y|X}[(Y - E_{Y|X}[y|x_i])^2|x_i] \\ &= \sum_j (y_j - E_{Y|X}[Y|x_i])^2 p_{Y|X}[y_j|x_i]\end{aligned}$$

Random Vectors
CDF, PMF, PDF

JKU

Joint Probability Density Function

- *Let* $[X, Y]^T$ *be a continuous 2-dimensional random vector with CDF* $F_{\mathbf{X}}(\mathbf{x})$. *The function* $p_{\boldsymbol{X}}(\boldsymbol{x})$

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{X}}(x, y) = \frac{\partial^2 F_{\mathbf{X}}(x, y)}{\partial x \partial y}$$

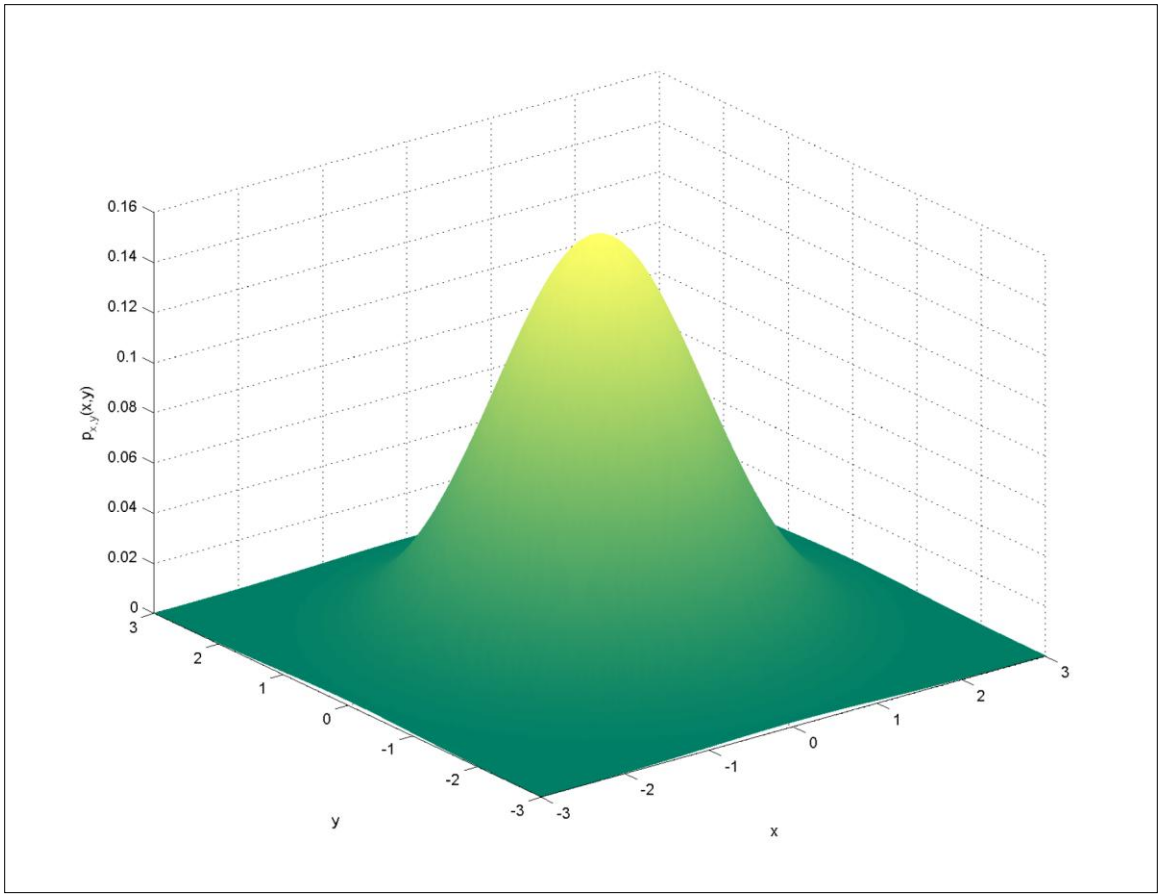
  is called the joint probability density function (joint PDF) of $\boldsymbol{x}$.

- $p_{\mathbf{X}}(\mathbf{x})\partial x \partial y$ can be regarded as a "probability".

- Properties:
  - $p_{\mathbf{X}}(x, y) \geq 0$
  - $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p_{\mathbf{X}}(x, y) dx dy = 1$

Random Vectors
CDF, PMF, PDF

- Graphical illustration of a standard bivariate Gaussian PDF for statistically independent X and Y:



Joint Probability Density Function

- Properties:
  - The probability of an event A is simply given by integrating $p_{\mathbf{X}}(x, y)$ over the range space corresponding to the event A.
  - Relationship between joint CDF and joint PDF:

$$F_{\mathbf{X}}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} p_{\mathbf{X}}(u, v) \mathrm{d}u \mathrm{d}v$$

Joint Probability Density Function

- Marginal PDFs:

$$p_X(x) = \int_{-\infty}^{\infty} p_{\mathbf{X}}(x, y)\mathrm{d}y \qquad p_Y(y) = \int_{-\infty}^{\infty} p_{\mathbf{X}}(x, y)\mathrm{d}x$$

- Two continuous random variables are defined to be statistically independent, if

$$p_{\mathbf{X}}(x, y) = p_X(x)p_Y(y).$$

- This is equivalent to

$$F_{\mathbf{X}}(x, y) = F_X(x)F_Y(y).$$

Conditional Probability Density Function

- The *conditional* PDF of Y, given that $X = x$, is

$$p_{Y|X}(y|x) = \frac{p_{\mathbf{X}}(x,y)}{p_X(x)} \qquad p_X(x) > 0$$

- The corresponding definition also holds for $p_{X|Y}(x|y)$.

- Notice that if X and Y are independent, then

$$p_{Y|X}(y|x) = p_Y(y).$$

- Conditional expectations are defined in the same way as for marginal PDFs.

Conditional Mean and Conditional Variance

- The *conditional mean* of Y, given that $X = x$, is defined as

$$\mu_{Y|X} = E_{Y|X}[Y|x] = \int_{-\infty}^{\infty} y p_{Y|X}(y|x)\mathrm{d}y$$

- Note that the conditional mean is a function of x.

- The conditional variance is defined as

$$\begin{aligned}\mathrm{var}[Y|x] &= E_{Y|X}[(Y - E_{Y|X}[Y|x])^2|x] \\ &= \int_{-\infty}^{\infty} (y - E_{Y|X}[Y|x])^2 p_{Y|X}(y|x)\mathrm{d}y\end{aligned}$$

Random Vectors
Expected Values

- To indicate the averaging PDF we will in many cases use corresponding indices in the expectation operator, e.g. $E_{X,Y}[X,Y]$.
- Let $Z = g(X,Y)$, then it can be shown that

$$E_Z[Z] = E_{X,Y}[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)p_{X,Y}(x,y)\mathrm{d}x\mathrm{d}y$$

- If $Z = g(X,Y)$=$\alpha X + \beta Y$, then it immediately follows that

$$E_{X,Y}[\alpha X + \beta Y] = \alpha E_X(X) + \beta E_Y[Y]$$ …Linearity

- Furthermore, we have

$$E_{X,Y}[g(X)] = E_X[g(X)]$$

- If X and Y are statistically independent, it can be shown that

$$E_{X,Y}[g(X)h(Y)] = E_X[g(X)]E_Y[h(Y)]$$

Random Vectors
Expected Values and Joint Moments

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- The *covariance* of two RVs is defined as

$$\mathrm{cov}(X,Y) = E_{X,Y}[(X - E_X[X])(Y - E_Y[Y])].$$

- The covariance is a measure of how the RVs *covary* with respect to each other. If they vary (on the mean) in the same direction, then the covariance will be positive, otherwise negative.

- The covariance has the following properties:

$$\mathrm{cov}(X,Y) = E_{X,Y}[XY] - E_X[X]E_Y[Y]$$
$$\mathrm{cov}(X,X) = \mathrm{var}(X)$$
$$\mathrm{cov}(Y,X) = \mathrm{cov}(X,Y)$$

Random Vectors
Expected Values and Joint Moments

- Two RVs X and Y are called *uncorrelated* if $\mathrm{cov}(X,Y) = 0$.
- Two RVs X and Y are called *orthogonal* if $E_{X,Y}[XY] = 0$.
- Consequently if at least one of the two random variables has zero mean, then orthogonality and uncorrelatedness are equivalent.
- If X and Y are statistically independent, then $\mathrm{cov}(X,Y) = 0$, since

$$\mathrm{cov}(X,Y) = E_{X,Y}[(X - E_X[X])(Y - E_Y[Y])]$$

$$= E_X[X - E_X[X]]E_Y[Y - E_Y[Y]]$$

$$= (E_X[X] - E_X[X])(E_Y[Y] - E_Y[Y]) = 0$$

The converse is not true in general.

Random Vectors
Expected Values and Joint Moments

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- The variance of a sum of RVs can be shown to be

$$\mathrm{var}[X+Y] = \mathrm{var}(X) + \mathrm{var}(Y) + 2\,\mathrm{cov}(X,Y).$$

  That means that the variance for uncorrelated RVs is

$$\mathrm{var}[X+Y] = \mathrm{var}(X) + \mathrm{var}(Y).$$

- The *correlation coefficient,* which measures the degree of statistical similarity between two random variables is defined as

$$\rho_{X,Y} = \frac{\mathrm{cov}(X,Y)}{\sqrt{\mathrm{var}(X)\,\mathrm{var}(Y)}}.$$

- It can easily be shown that $|\rho_{X,Y}| \leq 1$

- Furthermore $\rho_{X,Y} = \begin{cases} 1 & \text{if } Y = aX + b \text{ with } a > 0 \\ -1 & \text{if } Y = aX + b \text{ with } a < 0 \end{cases}.$

Random Vectors
Linear Prediction of a Random Variable Outcome

Example: We seek a linear (actually affine) predictor (or estimator) of Y based on knowledge of the outcome of X:

$$\hat{Y} = aX + b$$

(The „hat“ indicates an estimator“.)

- As a measure of closeness we choose the (Bayesian) mean square error (MSE)

$$J = E_{X,Y}\left[(Y - \hat{Y})^2\right].$$

- At the next slide we will show that the solution and therefore the optimal linear prediction of Y given the outcome $X = x$ is

$$\hat{Y} = E_Y[Y] + \frac{\mathrm{cov}(X,Y)}{\mathrm{var}(X)}(x - E_X[X])$$

- Guidance through the derivation:
  - Inserting the linear predictor into the cost function gives

$$J = E_{X,Y}\left[(Y - aX - b)^2\right]$$
$$= \dots$$
$$= E_Y[Y^2] - 2aE_{X,Y}[XY] + a^2E_X[X^2] - 2bE_Y[Y] + 2abE_X[X] + b^2$$

  - Calculating the gradient and setting it to the zero vector immediately leads to the optimal coefficients

$$a_{opt} = \frac{E_{X,Y}[XY] - E_X[X]E_Y[Y]}{E_X[X^2] - E_X^2[X]} = \frac{\mathrm{cov}(X,Y)}{\mathrm{var}(X)}$$
$$b_{opt} = E_Y[Y] - a_{opt}E_X[X] = E_Y[Y] - \frac{\mathrm{cov}(X,Y)}{\mathrm{var}(X)}E_X[X]$$

  - Inserting the coefficients into the linear predictor and setting $X = x$ finally leads to the solution given above.

Random Vectors
Linear Prediction of a Random Variable Outcome

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Remarks:

- Note that we refer to $\hat{Y} = aX + b$ as a *predictor* but $\hat{y} = ax + b$ as the *prediction* which is the value of the predictor.
- As expected, the prediction of Y based on $X = x$ depends on the covariance.
- If the covariance is zero, then

  $$\hat{Y} = E_Y[Y],$$

  in this case, X provides no information about Y.

- Example!

Random Vectors
Mean Vector and Covariance Matrix

Mean vector

- The *mean vector* of a random vector is defined by

$$\boldsymbol{\mu}_{\mathbf{X}} = E_{\mathbf{X}}[\mathbf{X}] = \begin{bmatrix} E_{X_1}[x_1] \\ \vdots \\ E_{X_n}[x_n] \end{bmatrix} = \begin{bmatrix} E_{\mathbf{X}}[x_1] \\ \vdots \\ E_{\mathbf{X}}[x_n] \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix}.$$

- The components of $\boldsymbol{\mu}_{\mathbf{X}}$ are the means of the individual random variables.

- Note that $E_{X_1}[x_1] = E_{\mathbf{X}}[x_1]$.

- Let $\mathbf{Y} = \mathbf{AX}$, then it easily follows that $\boxed{E_Y[\mathbf{Y}] = \mathbf{A}E_X[\mathbf{X}].}$

Random Vectors
Mean Vector and Covariance Matrix

Correlation and covariance matrices

- The *covariance matrix* of a random vector is defined by

$$\mathbf{C_{XX}} = E_{\mathbf{X}}[(\mathbf{X} - E_{\mathbf{X}}[\mathbf{X}])(\mathbf{X} - E_{\mathbf{X}}[\mathbf{X}])^T]$$

$$= \begin{bmatrix} \mathrm{var}(X_1) & \mathrm{cov}(X_1, X_2) & \cdots & \mathrm{cov}(X_1, X_n) \\ \mathrm{cov}(X_2, X_1) & \mathrm{var}(X_2) & \cdots & \mathrm{cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{cov}(X_n, X_1) & \mathrm{cov}(X_n, X_2) & & \mathrm{var}(X_n) \end{bmatrix}.$$

- If the elements of $\mathbf{X}$ are all uncorrelated to each other, then the covariance matrix becomes a diagonal matrix.

- The *correlation matrix* is defined by $\mathbf{R_{XX}} = E_{\mathbf{X}}[\mathbf{X}\mathbf{X}^T]$.

- We have that $\mathbf{C_{XX}} = \mathbf{R_{XX}} - \boldsymbol{\mu}_{\mathbf{X}}\boldsymbol{\mu}_{\mathbf{X}}^T$.

Random Vectors
Mean Vector and Covariance Matrix

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Properties of covariance matrices

It can be shown that

a) the covariance matrix is symmetric,

b) the covariance matrix is positive semidefinite,

c) the covariance matrix for uncorrelated random variables is diagonal,

d) the covariance matrix of $\mathbf{Y} = \mathbf{AX}$ is

$$\mathbf{C_{YY}} = \mathbf{A}\mathbf{C_{XX}}\mathbf{A}^T$$

e) the covariance matrix can always be diagonalized (by an appropriate linear transformation).

Random Vectors
Mean Vector and Covariance Matrix

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Correlation and covariance matrices

- The *n* x *m* *cross covariance matrix and cross correlation matrix* of two RVs of length *n* and *m* are defined by

$$\mathbf{C_{XY}} = E_{\mathbf{X,Y}}[(\mathbf{X} - E_{\mathbf{X}}[\mathbf{X}])(\mathbf{Y} - E_{\mathbf{Y}}[\mathbf{Y}])^T]$$

$$\mathbf{R_{XY}} = E_{\mathbf{X,Y}}[\mathbf{X}\mathbf{Y}^T].$$

- We have that $\mathbf{C_{XY}} = \mathbf{R_{XY}} - \boldsymbol{\mu}_{\mathbf{X}}\boldsymbol{\mu}_{\mathbf{Y}}^T.$

- Two random vectors $\mathbf{X}$ and $\mathbf{Y}$ are said to be
  - ☐ uncorrelated if: $\mathbf{C_{XY}} = \mathbf{0} \Rightarrow \mathbf{R_{XY}} = \boldsymbol{\mu}_{\mathbf{X}}\boldsymbol{\mu}_{\mathbf{Y}}^T$
  - ☐ orthogonal if: $\mathbf{R_{XY}} = \mathbf{0}$

Random Vectors
The Multivariate Gaussian Distribution

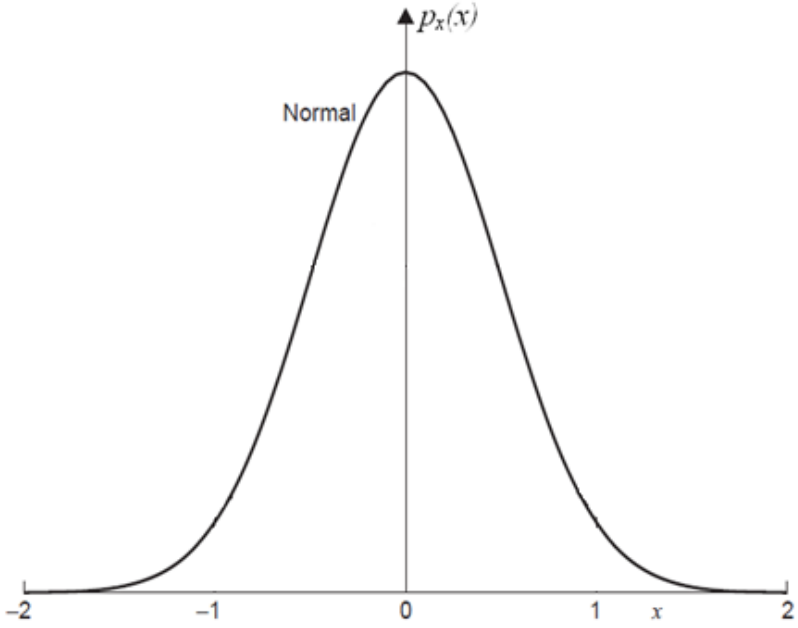
Recap: The Scalar Gaussian Distribution

- The Gaussian (or normal) distribution is by far the most important distribution in technical problems.

- The PDF of a Gaussian distributed random variable X with mean μ_X and variance σ_X^2 is denoted with $\mathcal{N}(\mu_X, \sigma_X^2)$ and given by

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2}$$

where $-\infty < \mu_X < \infty$ and $\sigma_X \geq 0$.



Definition of the Multivariate Gaussian Distribution

- *The multivariate Gaussian PDF is defined as*

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}\sqrt{\det \mathbf{C}_{\mathbf{XX}}}} \exp\left(-\frac{1}{2}[\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}}]^T \mathbf{C}_{\mathbf{XX}}^{-1}[\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}}]\right),$$

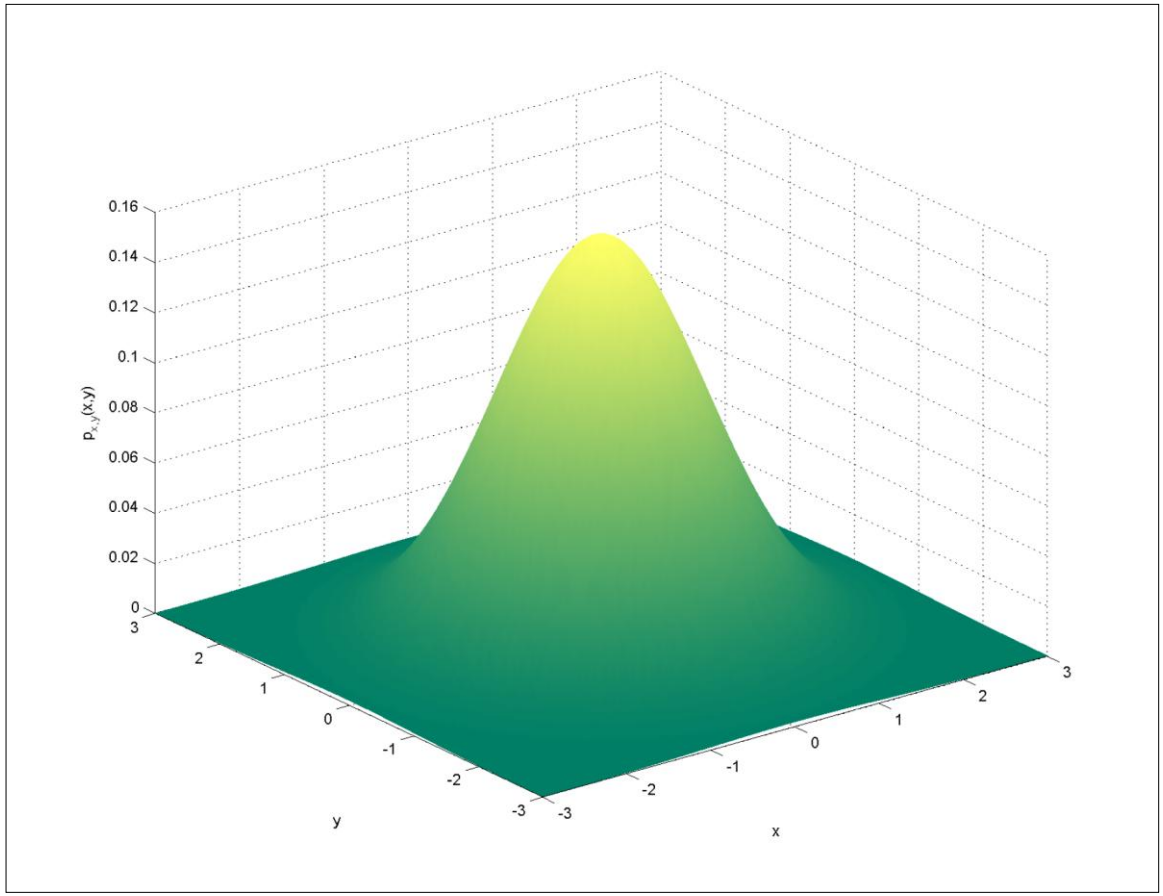
and we say that the elements of the random vector $\mathbf{X}$ *are jointly Gaussian distributed.*

- For $n = 2$ the distribution is also called *bivariate Gaussian distribution*.

- If $\boldsymbol{\mu}_{\mathbf{X}} = \mathbf{0}$ and $\mathbf{C}_{\mathbf{XX}} = \mathbf{I}$, the distribution is called *standard multivariate Gaussian distribution*.

Random Vectors
The Multivariate Gaussian Distribution

- Graphical illustration of a standard bivariate Gaussian PDF for statistically independent X and Y:



Properties of the Multivariate Gaussian Distribution

- For $n = 1$ the multivariate Gaussian distribution corresponds to the scalar Gaussian distribution.

- Every sub-vector of $\mathbf{X}$ also features a multivariate Gaussian distribution (or in case the sub-vector has length 1, a scalar Gaussian distribution), and the mean and covariance matrix are simply given by deleting appropriate entries of $\boldsymbol{\mu}_{\mathbf{X}}$ and appropriate columns and rows of $\mathbf{C}_{\mathbf{XX}}$.

- If $\mathbf{C}_{\mathbf{XX}}$ is a diagonal matrix which means that the elements of $\mathbf{X}$ are uncorrelated, then the multivariate Gaussian PDF can be written as

$$p_{\mathbf{X}}(\mathbf{x}) = p_{X_1}(x_1) \cdot p_{X_2}(x_2) \cdots p_{X_n}(x_n).$$

  That means for Gaussian RVs uncorrelatedness implies statistical independence.

Random Vectors
The Multivariate Gaussian Distribution

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Properties of the Multivariate Gaussian Distribution

Linear Transformation of Gaussian Random Vectors

If $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_\mathbf{X}, \mathbf{C}_{\mathbf{XX}})$ *and* $\mathbf{Z} = \mathbf{G}\mathbf{X}$ *with an invertible* $\mathbf{G}$, *then*

$$\mathbf{Z} \sim \mathcal{N}(\boldsymbol{\mu}_\mathbf{Z}, \mathbf{C}_{\mathbf{ZZ}})$$

with

$$\boldsymbol{\mu}_\mathbf{Z} = \mathbf{G}\boldsymbol{\mu}_\mathbf{X}$$

$$\mathbf{C}_{\mathbf{ZZ}} = \mathbf{G}\mathbf{C}_{\mathbf{XX}}\mathbf{G}^T.$$

That means a linear transformation of a Gaussian random vector with an invertible $\mathbf{G}$ *results in another Gaussian random vector.*

Random Vectors
The Multivariate Gaussian Distribution

Conditional PDF of Multivariate Gaussian RV

Let's consider the stacked random vector $\left[\mathbf{X}^T, \mathbf{Y}^T\right]^T$ *where* $\mathbf{X}$ *is a* $k \times 1$ *vector and* $\mathbf{Y}$ *is an* $l \times 1$ *vector, and* $\mathbf{X}$ *and* $\mathbf{Y}$ *are jointly Gaussian, with mean vector and partitioned covariance matrix in the form*

$$\begin{bmatrix} E[\mathbf{X}] \\ E[\mathbf{Y}] \end{bmatrix}; \mathbf{C} = \begin{bmatrix} \mathbf{C_{XX}} & \mathbf{C_{XY}} \\ \mathbf{C_{YX}} & \mathbf{C_{YY}} \end{bmatrix}$$

so that $$p_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = \frac{1}{(2\pi)^{\frac{k+l}{2}}\sqrt{\det \mathbf{C}}} \exp\left(-\frac{1}{2}\begin{bmatrix} \mathbf{x} - E[\mathbf{X}] \\ \mathbf{y} - E[\mathbf{Y}] \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} \mathbf{x} - E[\mathbf{X}] \\ \mathbf{y} - E[\mathbf{Y}] \end{bmatrix}\right)$$

then the conditional PDF $p_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x})$ *is also Gaussian and*

$$E[\mathbf{Y}|\mathbf{x}] = E[\mathbf{Y}] + \mathbf{C_{YX}}\mathbf{C_{XX}^{-1}}(\mathbf{x} - E[\mathbf{X}])$$

$$\mathbf{C_{Y|X}} = \mathbf{C_{YY}} - \mathbf{C_{YX}}\mathbf{C_{XX}^{-1}}\mathbf{C_{XY}}.$$

Random Vectors
The Multivariate Gaussian Distribution

The Bivariate Gaussian Distribution

- We now concentrate on the bivariate Gaussian distribution

$$p_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{\det(\mathbf{C})}} \exp\left(-\frac{1}{2}\begin{bmatrix} x-\mu_X \\ y-\mu_Y \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} x-\mu_X \\ y-\mu_Y \end{bmatrix}\right)$$

with covariance matrix

$$\mathbf{C} = \begin{bmatrix} \sigma_X^2 & \mathrm{cov}(X,Y) \\ \mathrm{cov}(Y,X) & \sigma_Y^2 \end{bmatrix} = \begin{bmatrix} \sigma_X^2 & \rho_{X,Y}\sigma_X\sigma_Y \\ \rho_{X,Y}\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix}.$$

- The marginal PDFs are $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$.

The Bivariate Gaussian Distribution

- The conditional PDF $p_{Y|X}(y|x)$ is also Gaussian with a new mean and a new variance as

$$E[Y|x] = E[Y] + \frac{\text{cov}(X,Y)}{\text{var}(X)}(x - E[X])$$

$$\text{var}(Y|x) = \text{var}(Y) - \frac{\text{cov}^2(X,Y)}{\text{var}(X)}$$

- Assuming that X and Y are not statistically independent the conditional PDF becomes more concentrated since there is less uncertainty about Y. To verify this, note that

$$\text{var}(Y|x) = \text{var}(Y)\left(1 - \frac{\text{cov}^2(X,Y)}{\text{var}(X)\,\text{var}(Y)}\right) = \text{var}(Y)\left(1 - \rho_{X,Y}^2\right).$$

The Bivariate Gaussian Distribution

- For $\mu_X = \mu_Y = 0$ and $\sigma_X^2 = \sigma_Y^2 = 1$ it can be shown, that the PDF can also be written in the form

$$p_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho_{X,Y}^2}} \exp\left[-\frac{1}{2(1-\rho_{X,Y}^2)}\left(x^2 - 2\rho_{X,Y}xy + y^2\right)\right].$$

- If X and Y are also uncorrelated (standard bivariate Gaussian) then

$$p_{X,Y}(x,y) = \frac{1}{2\pi}\exp\left[-\frac{1}{2}\left(x^2+y^2\right)\right]$$

$$= \underbrace{\frac{1}{\sqrt{2\pi}}\exp\left[-\frac{1}{2}x^2\right]}_{p_X(x)} \underbrace{\frac{1}{\sqrt{2\pi}}\exp\left[-\frac{1}{2}y^2\right]}_{p_Y(y)} = p_X(x)p_Y(y)$$

The Central Limit Theorem

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- **Theorem**: *If $X_1, X_2, \ldots, X_n$ are continuous independent and identically distributed (IID) random variables, each with mean $E[X]$ and variance $\mathrm{var}(X)$, then for the sum $S_n = \sum_{i=1}^{n} X_n$ of those RVs as $n \to \infty$*

$$\frac{S_n - E[S_n]}{\sqrt{\mathrm{var}(S_n)}} = \frac{\sum_{i=1}^{n} X_n - nE[X]}{\sqrt{N\mathrm{var}(X)}} \to \mathcal{N}(0,1).$$

- This is the famous central limit theorem, which says that the PDF of the standardized sum of a large number of continuous IID random variables will converge to a Gaussian PDF.

- This important theorem holds for arbitrary distributions of X_i (with finite mean and finite variance).

The Central Limit Theorem

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- Several other versions of the CLT exist, each applying in the context of different conditions.
- Frequently versions of the CLT are applied to sums of independent (but not necessarily identically distributed) random variables, where the limiting distribution is a normal distribution with a non-standard mean and variance (i.e., not necessarily 0 and 1).
- In many engineering applications, such as noise modeling, the total noise can be represented as the sum of many independent sources. In such cases, it is reasonable to model the noise samples as being distributed according to $\mathcal{N}(\mu, \sigma^2)$.